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W101-001 REV A

REVISIONS

ZONE	REV	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	2026-10-08	J. SMITH

NOTES:

1. INTERPRET DRAWING PER ASME Y14.100.

2. DIMENSIONS ARE IN INCHES. LENGTH TOLERANCE ±0.5 IN.

3. WIRE PER WIRE LIST. TERMINATE CONTACTS PER CONNECTOR MANUFACTURER'S INSTRUCTIONS.

4. 100% CONTINUITY AND ISOLATION TEST PER WIRE LIST.

5. ASSEMBLY VIEW AND CONNECTOR PINOUTS ON SHEET 2. WIRING DIAGRAM ON SHEET 3. WIRE LIST, WIRE GROUPS, SPLICES AND LABEL SCHEDULE ON SHEET 4.

6. DIMENSIONS PER IPC-D-620: HARNESS DIMENSIONS, INCLUDING BREAKOUT AND SPLICE LOCATIONS, ARE MEASURED FROM DATUM A, THE P1 CONNECTOR FACE. HARNESS LENGTH IS MEASURED FROM DATUM A TO THE FINAL TERMINATION. BREAKOUT LENGTHS ARE MEASURED FROM THE APPROXIMATE CENTERLINE OF THE HARNESS AT THE BREAKOUT TO THE FINAL TERMINATION.

7. CONTACTS (M39029/56-351, M39029/58-363) ARE SUPPLIED WITH CONNECTORS (FIND NOS. 1, 2); DO NOT ORDER SEPARATELY.

8. CONNECTOR PINOUTS SHOW THE FRONT (ENGAGING) FACE OF THE CONNECTOR CALLED OUT: INSERT ARRANGEMENT PER MIL-STD-1560, SOCKET INSERTS AS THE MIRROR IMAGE OF THE PIN INSERT. FILLED CAVITIES ARE WIRED; OPEN CAVITIES ARE NC. MASTER KEYWAY NOT SHOWN.

9. NC (NO CONNECTION) CONTACT POSITIONS: P1: N, P, R, S, T, U; P2: N, P, R, S, T, U.

10. TWIST THE WIRES OF EACH TWISTED GROUP TOGETHER OVER THEIR FULL LENGTH (SEE WIRE GROUPS TABLE).

11. CALCULATED BUNDLE DIAMETER AT CONNECTORS (REF): P1 0.257 IN, P2 0.257 IN.

12

TERMINATE SHIELDS PER WIRE GROUPS TABLE USING FIND NO. 13. INSULATE FLOATING SHIELD ENDS.

13

INSTALL SPLICE (FIND NO. 14) AT LOCATION SHOWN; SEE SPLICE TABLE.

14

RECOVER HEATSHRINK BOOT (FIND NO. 4) OVER BACKSHELL CABLE CLAMP.

15

INSTALL IDENTIFICATION LABEL (FIND NO. 5) ADJACENT TO CONNECTOR; MARK PER LABEL SCHEDULE.

16

INSTALL WIRE MARKER (FIND NO. 15) AT EACH END OF EACH WIRE; MARK WITH WIRE ID.

15	26		TMS-SCE-1/8-2.0-9	MARKER, WIRE, 1/8 IN	16
14	1		M81824/1-1	SPLICE, CRIMP, ENVIRONMENT RESISTANT	13
13	2		M83519/2-3	SHIELD TERMINATION, SOLDER SLEEVE	12
12	48 IN		M22759/16-20-5	WIRE, 20 AWG, GRN	
11	54 IN		M22759/16-22-0	WIRE, 22 AWG, BLK	
10	48 IN		M22759/16-22-6	WIRE, 22 AWG, BLU	
9	AR		M22759/16-22-9	WIRE, 22 AWG, WHT	
8	48 IN		M22759/16-20-0	WIRE, 20 AWG, BLK	
7	48 IN		M22759/16-20-2	WIRE, 20 AWG, RED	
6	48 IN		M27500-22TG2T14	CABLE, SHIELDED TWISTED PAIR, 22 AWG	
5	2		TMS-SCE-1/2-2.0-9	SLEEVE, IDENTIFICATION, 1/2 IN	15
4	2		202K153-25/225-0	BOOT, HEATSHRINK, STRAIGHT	14
3	2		M85049/38S15W	BACKSHELL, STRAIGHT, SHELL 15	
2	1		D38999/26WD18PN	CONNECTOR, PLUG, 18 PIN	
1	1		D38999/26WD18SN	CONNECTOR, PLUG, 18 SKT	
FIND NO	QTY REQD	CAGE CODE	PART OR IDENTIFYING NO	NOMENCLATURE OR DESCRIPTION	NOTE

PARTS LIST

APPLICATION		UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES LENGTH TOLERANCE ±0.5 INTERPRET PER ASME Y14.5 DO NOT SCALE DRAWING	CONTRACT NO.		ACME AEROSPACE (EXAMPLE)						
NEXT ASSY	USED ON		APPROVALS DATE		TITLE						
A1000-001	FCC INST...		DRAWN K. KIPER		2026-10-08		CABLE ASSY, FCC TO PDU				
			CHECKED J. SMITH		2026-10-08						
		THIRD ANGLE PROJECTION		ENGR				SIZE C	CAGE CODE	DWG NO. W101-001	REV A
				APPROVED				SCALE NONE		WEIGHT	SHEET 1 OF 4

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W101-001 REV A

ASSEMBLY VIEW AND CONNECTOR PINOUTS (NOT TO SCALE)

C

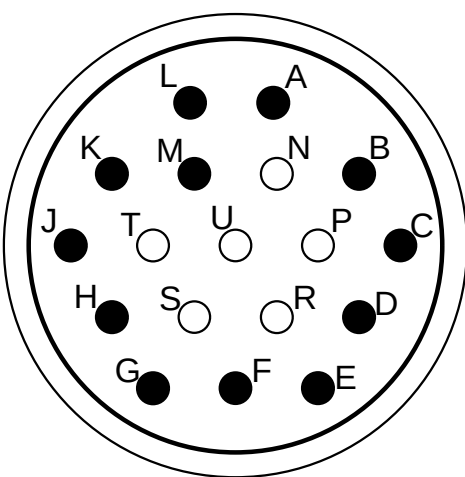
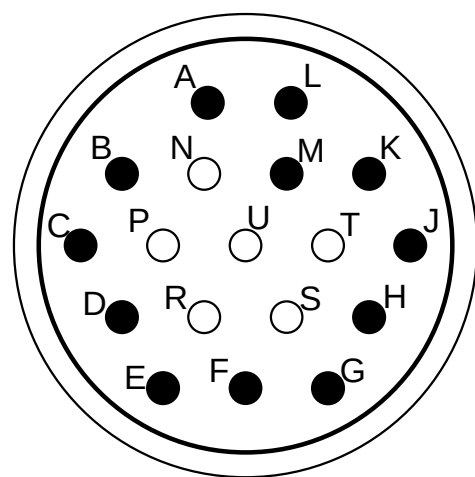
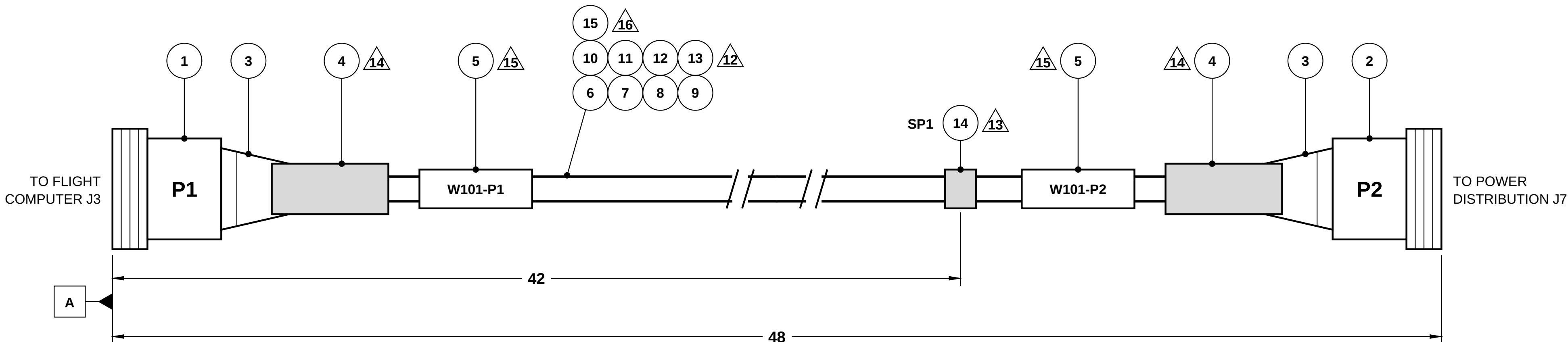
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SIZE	CAGE CODE	DWG NO.	REV
C		W101-001	A
SCALE	WEIGHT	SHEET	
NONE		2 OF 4	

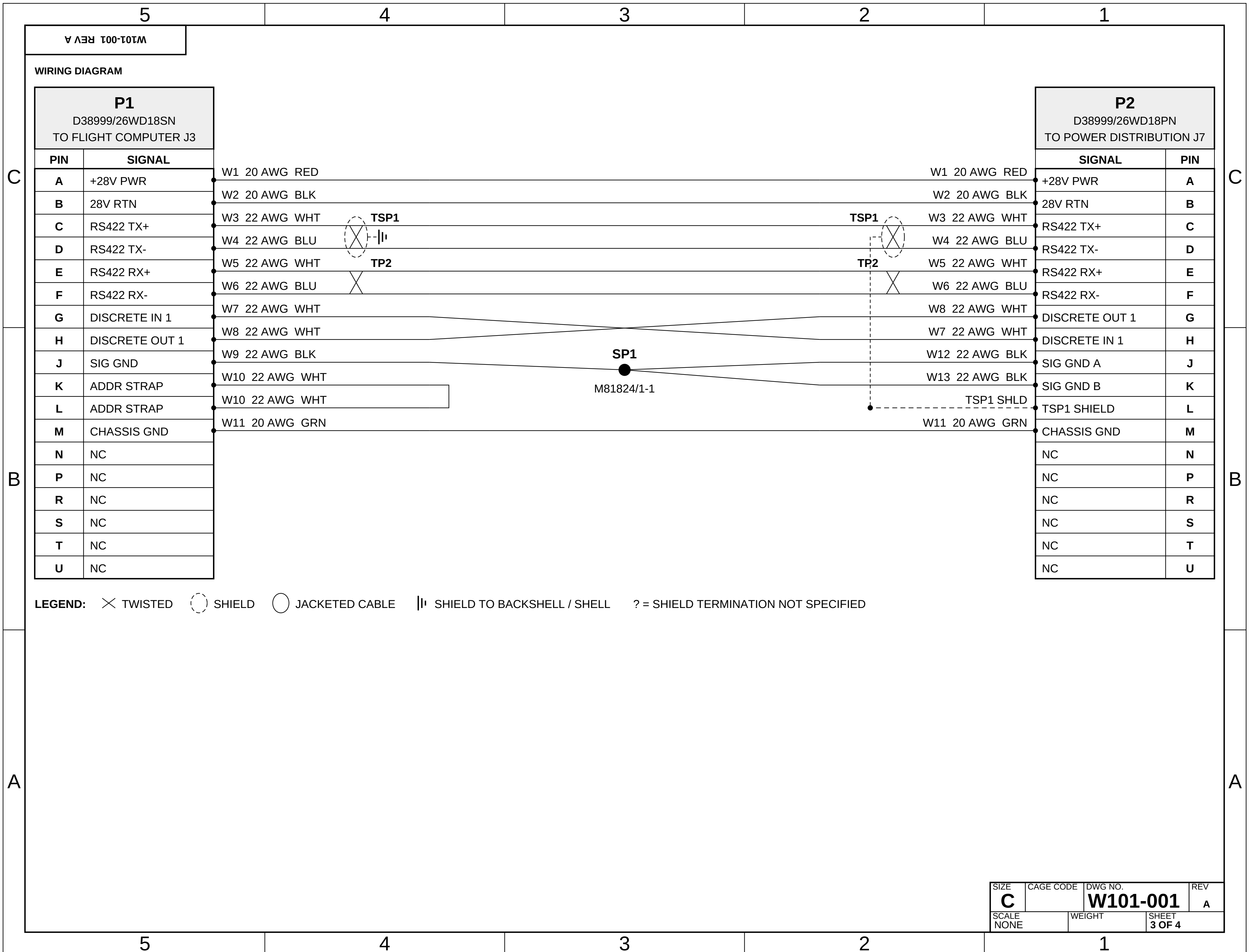
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W101-001 REV A																					
WIRE LIST AND TABLES																					
WIRE LIST																					
WIRE	FROM	PIN	TO	PIN	SIGNAL	AWG	COLOR	WIRE P/N	GROUP	LENGTH (IN)	NOTES										
W1	P1	A	P2	A	+28V PWR	20	RED	M22759/16-20-2		48											
W2	P1	B	P2	B	28V RTN	20	BLK	M22759/16-20-0		48											
W3	P1	C	P2	C	RS422 TX+	22	WHT	(M27500-22TG2T14)	TSP1	48											
W4	P1	D	P2	D	RS422 TX-	22	BLU	(M27500-22TG2T14)	TSP1	48											
W5	P1	E	P2	E	RS422 RX+	22	WHT	M22759/16-22-9	TP2	48											
W6	P1	F	P2	F	RS422 RX-	22	BLU	M22759/16-22-6	TP2	48											
W7	P1	G	P2	H	DISCRETE IN 1	22	WHT	M22759/16-22-9		48											
W8	P1	H	P2	G	DISCRETE OUT 1	22	WHT	M22759/16-22-9		48											
W9	P1	J	SP1		SIG GND	22	BLK	M22759/16-22-0		42											
W10	P1	K	P1	L	ADDR STRAP	22	WHT	M22759/16-22-9		AR	JUMPER IN P1										
W11	P1	M	P2	M	CHASSIS GND	20	GRN	M22759/16-20-5		48											
W12	SP1		P2	J	SIG GND A	22	BLK	M22759/16-22-0		6											
W13	SP1		P2	K	SIG GND B	22	BLK	M22759/16-22-0		6											
WIRE GROUPS AND SHIELDS																					
GROUP	TYPE				WITHIN	WIRES	CABLE P/N	SHIELD P/N	SHIELD TERM	SHIELD (FROM END)	SHIELD (TO END)	NOTES									
TSP1	SHIELDED TWISTED PAIR					W3, W4	M27500-22TG2T14		M83519/2-3 (FIND NO. 13)	P1: BACKSHELL	P2: PIN L	RS422 TX									
TP2	TWISTED PAIR					W5, W6						RS422 RX; 10 TWISTS PER FOO...									
SPLICES																					
SPLICE	FIND NO	SPLICE P/N			LOCATION			WIRES		NOTES											
SP1	14	M81824/1-1			42 IN FROM DATUM A (P1 FACE)			W9, W12, W13		SIGNAL GROUND SPLICE											
LABEL SCHEDULE																					
FIND NO	LABEL P/N			LEGEND	LOCATION				QTY												
5	TMS-SCE-1/2-2.0-9			W101-P1	CABLE, ADJACENT TO P1				1												
5	TMS-SCE-1/2-2.0-9			W101-P2	CABLE, ADJACENT TO P2				1												
15	TMS-SCE-1/8-2.0-9			W1	WIRE W1, AT P1-A AND P2-A				2												
15	TMS-SCE-1/8-2.0-9			W2	WIRE W2, AT P1-B AND P2-B				2												
15	TMS-SCE-1/8-2.0-9			W3	WIRE W3, AT P1-C AND P2-C				2												
15	TMS-SCE-1/8-2.0-9			W4	WIRE W4, AT P1-D AND P2-D				2												
15	TMS-SCE-1/8-2.0-9			W5	WIRE W5, AT P1-E AND P2-E				2												
15	TMS-SCE-1/8-2.0-9			W6	WIRE W6, AT P1-F AND P2-F				2												
15	TMS-SCE-1/8-2.0-9			W7	WIRE W7, AT P1-G AND P2-H				2												
15	TMS-SCE-1/8-2.0-9			W8	WIRE W8, AT P1-H AND P2-G				2												
15	TMS-SCE-1/8-2.0-9			W9	WIRE W9, AT P1-J AND SP1				2												
15	TMS-SCE-1/8-2.0-9			W10	WIRE W10, AT P1-K AND P1-L				2												
15	TMS-SCE-1/8-2.0-9			W11	WIRE W11, AT P1-M AND P2-M				2												
15	TMS-SCE-1/8-2.0-9			W12	WIRE W12, AT SP1 AND P2-J				2												
15	TMS-SCE-1/8-2.0-9			W13	WIRE W13, AT SP1 AND P2-K				2												